Day 4: Mixed Masks (/18 to /30)

Level 1, Month 1, Networking foundations

What I learned

- Between which two boundaries does the mask fall? That decides which number carries the blocks:

Mask	Number with the blocks	Boundary
/17 to /23	Third	24
/25 to /30	Fourth	32

- When the blocks are in the fourth number, the first three numbers stay exactly as in the given address.
- End of the block = start of the NEXT block - 1. Never start of my own block - 1.
- Use the block size from the table (4, 8, 16, 32...), not the number of bits (2, 3, 4, 5...).
- Valid hosts = 2^(bits d'hote) - 2.

Self-checks that catch mistakes before sending:

- The given address must lie between reseau and broadcast.
- Broadcast must be greater than reseau, and derniere greater than premiere.
- The first three numbers must stay the same when the blocks are in the fourth number.

Problems solved

Address	Bits d'hote	Taille du bloc	Reseau	Premiere	Broadcast
172.20.133.45/22	10	4	172.20.132.0	172.20.132.1	172.20.135.255
10.50.77.200/19	13	32	10.50.64.0	10.50.64.1	10.50.95.255
192.168.200.99/25	7	128	192.168.200.0	192.168.200.1	192.168.200.127
192.168.1.9/30	2	4	192.168.1.8	192.168.1.9	192.168.1.11

Valid hosts: /19 gives 8190 (2^13 - 2), and /30 gives 2 (2^2 - 2).

Error log

Mistake	Correct answer	Why
172.20.133.45/22: broadcast .131.255	.135.255	I subtracted 1 from the start of my own block (132). It must come from the next block (136).
Used 2 as the step	4	2 is the bits in the third number. The step is the block size, 4.
Valid hosts in /19: 19	8190	I used the mask. The rule is 2^13 - 2.
192.168.200.99/25: put the block 128 in the third number	Blocks are in the fourth number	/25 is between boundaries 24 and 32, so the first three numbers stay fixed.
192.168.1.9/30: block .0 to .3	Block .8 to .11	The given address must be inside the block.
Broadcast .7, derniere .6	.11 and .10	I took 8 - 1. It must be the next block start, 12, minus 1.
Valid hosts in /30: 14	2	2^2 = 4, then 4 - 2 = 2. I used the wrong exponent.

Main lessons

- The mask tells me which number carries the blocks. Decide that first.
- Broadcast always comes from the next block start minus 1.
- Self-check the order: given address inside the block, derniere right before broadcast.
- I fixed one mistake on my own today (I noticed that I used 2 instead of 4).
- The 10.50.77.200/19 problem was fully correct on the first try.

Anki

Cards for the block-start rule (next block start - 1) and the self-checks.

Next session

A first look at a pcap file in Wireshark, then a short subnetting review.